

LEARNING COMPANION

How this case study was built

Every technique in the Fungi Labs appraisal, explained — and what to be sceptical about

Particulars	Detail
Purpose	To explain the reasoning behind each analysis, not just its output
Audience	Anyone learning project finance, financial modelling or healthcare appraisal
Covers	Model architecture, revenue and cost build, debt structuring, DSCR, IRR, NPV, sensitivity, tornado, Monte Carlo, valuation, cap tables
Companion to	Feasibility Study, Detailed Project Report, Viability & Bankability Report, master model workbook
How to use it	Read alongside the workbook. Section 3 is a guided tour of all 27 sheets.
Document	Learning Companion — 4 of 4
Date	8 August 2026

Basis of preparation and limitations

This document explains method. It is not a textbook and it is not authoritative on Indian accounting, tax or banking regulation — verify anything you intend to rely on against primary sources.

Where a technique has a known weakness, this companion says so rather than presenting it as settled. That includes the techniques used in this very case study: Section 10 is a list of things the model does badly or not at all.

Contents

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1. What an appraisal set is for

Four documents accompany this model, and they are not four versions of the same thing. Each answers a different question for a different reader, and the most common mistake in practice is to write one document and label it four ways.

Document	Answers	Written for	Characteristic mistake
Feasibility Study	Should this be built at all?	The promoter, before committing	Becoming a sales document — only the base case, no falsification test
Detailed Project Report	What exactly is being built, and how is it funded?	The lender's appraisal team	Burying the risks so the credit committee finds them instead
Viability & Bankability Report	Will the debt be repaid, and under what conditions?	The credit committee	Confusing "the asset is good" with "the loan is safe"
Learning Companion	Why was it analysed this way?	Whoever inherits the model	Not existing — most models arrive with no explanation at all

The distinction between the second and third documents is the one most often collapsed, and it matters. A DPR describes and requests. A viability report forms an opinion, including an opinion the borrower may not like. In this set the viability report recommends de-levering the deal against the promoter's stated preference, and says so plainly. That is what makes it useful.

1.1 The single most important idea in this case study

An asset can be sound while its financing is fragile

This project produces a positive net present value in 91% of simulated futures and a technical default in 27% of them. Both numbers are correct, and they are not in tension.

Net present value asks whether the asset is worth building across its whole life. DSCR asks whether cash arrives in the specific years a loan agreement demands it. A hospital can be an excellent twenty-year asset and still miss a payment in Year 5, because the loan does not care that the money shows up in Year 8.

Almost every technique in this model exists to keep those two questions apart. If you take one thing from this companion, take that: value and liquidity are different problems, they have different solutions, and conflating them is how good projects get financed badly.

2. Model architecture

2.1 Build drivers, never growth rates

The weakest thing a financial model can do is start with a revenue number and grow it by a percentage. It hides every operational assumption inside one figure, so nobody can challenge it and nothing can be tested. This model never does that. Inpatient revenue is built as:

The revenue identity

Inpatient revenue = Beds × 365 × Occupancy × Net ARPOB

Net ARPOB = Gross tariff ARPOB × Weighted realisation index

Admissions = Occupied bed-days ÷ Average length of stay

Every term is separately visible, separately challengeable and separately sensitisable. When the tornado analysis reports that peak occupancy swings equity IRR by 11.3%, that is only possible because occupancy exists as its own input rather than being buried in a growth assumption.

Outpatient revenue follows the same discipline: footfall per day multiplied by 365 multiplied by revenue per visit. Diagnostics and pharmacy are driven off outpatient footfall through a conversion rate and a ticket size. Dialysis is stations multiplied by shifts multiplied by operating days multiplied by utilisation multiplied by price. Nothing is a percentage of the year before.

2.2 The calculation chain

A model should flow in one direction, so that any number can be traced to its inputs without circular references. This one runs:

Stage	Sheets	What it establishes
1. Switches	Control Panel	Scenario, funding structure and payer mix — three cells that drive everything
2. Inputs	Assumptions	Every operating, cost, working-capital, tax and valuation assumption in one place
3. Investment	Capex & Funding, Depreciation	What is spent, how it is funded, how it is written off
4. Operations	Revenue Build, Manpower, Operating Costs	Revenue and cost, built from drivers
5. Statements	P&L, Cash Flow, Balance Sheet, Working Capital	The three statements, with a balance check
6. Financing	Repayment Schedule, Debt Schedule	Drawdown, moratorium, amortisation, interest
7. Tests	DSCR & Covenants, Break-even, Ratio Dashboard	Can it service the debt, and at what occupancy
8. Returns	Returns & IRR, Valuation, Unit Economics	What the asset and the shareholder earn
9. Risk	Scenarios, Sensitivity, Tornado, Monte Carlo, Funding Comparison	What happens when the assumptions are wrong

2.3 Three checks worth building into any model

These take minutes to add and catch the errors that otherwise survive to the credit committee:

1. A balance sheet check. Total assets minus total liabilities, in every year, with an OK/ERROR flag. If it does not read zero, something is wrong — and it is almost always the cash line or a sign error in the cash flow. In this model the check reads zero to ten decimal places in all ten years.
2. Additivity checks. Bed configuration must sum to the bed count. The payer mix must sum to 100%. The repayment ladder must sum to 100% of the sanction. Means of finance must equal cost of project. Each is one formula, and each catches a class of error that is invisible otherwise.
3. A cross-check between two methods. This model computes DSCR two ways — on a cash basis using CFADS, and on the accrual basis most Indian sanction letters use. If they diverge unexpectedly, one of them is wrong.

A related discipline: verify against an independent implementation. Every number in this workbook was cross-checked against a separate Python implementation of the same model. That process found three genuine errors — a double-subtraction in the capex allocation base, a mismatch between the salary assumption and the manpower build, and a definitional error in exit equity value. All three produced plausible-looking output and none would have been caught by inspection.

2.4 Colour convention

Colour	Meaning	Why it matters
Blue text on yellow fill	Hardcoded input or scenario lever	The only cells anyone should type into
Black text	Formula computed on this sheet	Overwriting one breaks the model silently
Green text	Link to another sheet	Tells a reviewer where to look for the source

This convention is near-universal in professional modelling and it exists for one reason: a model is read far more often than it is built, usually by someone who did not build it. Anyone opening this workbook can see in a glance which cells are assumptions and which are consequences.

3. Guided tour of the workbook

Twenty-seven sheets, in the order they should be read.

Sheet	What to look at	What it teaches
Control Panel	The three switches at the top. Change one and the whole workbook recalculates.	Scenario architecture — how to make a model answer many questions rather than one
Assumptions	Column E, where every figure carries a source note or a reasoning note	Documenting assumptions where the reader will actually see them
Capex & Funding	The closed-form IDC solution, and the two views of gearing	Solving circularity algebraically; the difference between headline and real leverage
Depreciation	Straight-line book depreciation alongside written-down-value tax depreciation	Why book and tax depreciation differ, and what a deferred tax liability is
Revenue Build	The bed-days × ARPOB chain, and the revenue mix rows below it	Driver-based revenue modelling
Manpower	The cadre table, and salary cost as a percentage of revenue	How a fixed cost base behaves during a ramp
Operating Costs	The variable / fixed split and the contribution margin ratio	Where operating leverage comes from
Working Capital	Debtor days driven by the payer mix switch	Why payer mix is a balance-sheet decision, not just a pricing one
Repayment Schedule	All 46 quarters: 6 drawdown, 12 moratorium, 28 instalments	How a real term loan is actually structured
Debt Schedule	Interest computed on the average balance, not the opening balance	A small convention that materially changes the interest figure
P&L	The loss carry-forward block and the effective tax rate line	Why a profitable company can pay no tax for years
Cash Flow	FCFF and FCFE side by side	The two free cash flow definitions and when to use each
Balance Sheet	The check row at the bottom	Non-negotiable model hygiene
DSCR & Covenants	The disclosed TOL/TNW exception	How to raise a covenant you cannot meet, before the lender finds it
Returns & IRR	The reconciliation at the foot explaining why equity IRR exceeds project IRR	What leverage does to returns, in both directions
Break-even	Cash break-even occupancy and the margin of safety	The single most useful operating number in the model
Unit Economics	Per bed, per bed-day, per admission, per employee	The language hospital operators actually use
Ratio Dashboard	The DuPont decomposition at the foot	Breaking return on equity into margin, turnover and leverage
Valuation (DCF)	The terminal value share of enterprise value	Why most DCFs are mostly assumption
Scenario Summary	The asymmetry between upside and downside	Why fixed-cost businesses fail rather than underperform
Sensitivity	Nine two-way grids with conditional formatting	Reading a surface rather than a point
Tornado	The ranking, not the values	Where management attention actually belongs
Monte Carlo	Section C — the probabilities	The difference between "could" and "how likely"

Sheet	What to look at	What it teaches
Funding Comparison	Median IRR barely moves; default probability halves	The central structuring insight of the case
PE Case	Cap table at cost, and the exit waterfall	Why greenfield equity rounds have no step-up
Charts	Occupancy against break-even occupancy	One picture that contains the whole risk story

4. The techniques, explained

4.1 ARPOB and why it is the metric that matters

Average revenue per occupied bed-day is the central productivity measure in hospital finance. It is total inpatient revenue divided by occupied bed-days, and by convention it already includes drugs and diagnostics consumed by inpatients — which is why this model adds diagnostics and pharmacy revenue only for walk-in and referral business, to avoid double-counting.

Two subtleties are worth understanding. First, ARPOB rises with case-mix complexity, so an increase can mean the hospital is treating sicker patients rather than charging more. Second, it moves inversely with length of stay for the same episode: discharging a patient a day earlier raises ARPOB while reducing total revenue from that patient. A rising ARPOB is therefore not automatically good news, and a falling one is not automatically bad.

This model separates gross tariff ARPOB from net realised ARPOB through the realisation index, which is where payer mix enters:

Payer segment	Share of revenue	Realisation vs list tariff	Debtor days
Cash / self-pay	40%	100%	3
Private insurance / TPA	35%	94%	55
Government schemes (PMJAY, CGHS, ESI)	25%	72%	110
Weighted average	100%	0.9090	48.0

Why payer mix is the most underrated variable in Indian hospital finance

Government scheme volume looks like free occupancy. It is paid for twice.

Once in realisation: PMJAY and similar schemes settle at roughly 72% of list tariff, so the same bed-day earns 28% less. And again in working capital: those receivables run around 110 days against three days for a cash patient.

Switch the Control Panel to a government-heavy mix and watch three things move together — the realisation index falls from 0.909 to 0.859, debtor days rise from 48.0 to 64.2, and net working capital rises by roughly Rs 3.97 Cr. Minimum DSCR falls from 1.46x to 1.21x.

This is why the payer mix is built as a switch rather than a fixed assumption. It is not a pricing decision — it is simultaneously a pricing, working-capital and covenant decision, and modelling it as one number hides two of those three.

4.2 Operating leverage — the engine and the risk

Separate costs into those that vary with revenue and those that do not, and a hospital's behaviour becomes obvious. Variable costs here are the consultant fee-for-service payout and consumables. Everything else — salaries, power, outsourced services, maintenance, administration, marketing, insurance — is fixed or semi-fixed.

Concept	Formula	This project (Year 5)
Contribution margin ratio	$(\text{Revenue} - \text{Variable costs}) \div \text{Revenue}$	67.0%
Operating break-even revenue	$\text{Fixed costs} \div \text{Contribution margin ratio}$	Rs 4,556 L of fixed cost to cover
Cash break-even occupancy	$\text{Occupancy} \times \text{Cash break-even revenue} \div \text{Actual revenue}$	61.0%
Margin of safety	$\text{Actual occupancy} - \text{Cash break-even occupancy}$	9.0%

Concept	Formula	This project (Year 5)
Degree of operating leverage	Contribution ÷ EBIT	3.9x

A contribution margin ratio of about 67% means roughly two-thirds of every additional rupee of revenue reaches EBITDA once fixed costs are covered. That is why occupancy dominates every sensitivity in this model. It is also why the downside is so severe: the same ratio applies in reverse, and fixed costs do not fall when the beds empty.

Cash break-even occupancy deserves particular attention because it is the number a hospital board should monitor above all others. It is the occupancy at which the business generates just enough to cover fixed costs, interest and principal. Below it, the hospital is consuming capital regardless of what the profit and loss account reports.

4.3 Structuring debt around a ramp

This is the most transferable lesson in the whole case study, and it generalises well beyond hospitals to any asset with a multi-year ramp — hotels, toll roads, schools, manufacturing plants.

Structure	Gearing	First principal instalment	Min DSCR	Breach years	Verdict
As originally proposed	70:30	Operating Year 2	0.27x	2	Not bankable
Recommended	70:30	Operating Year 4	1.46x	0	Bankable

Identical leverage. Identical pricing. Opposite outcomes.

The only difference between those two rows is when the first instalment falls due.

The reason is that cash break-even occupancy in the stabilised years is about 61%, and this hospital only crosses it during Year 3. Requiring amortisation in Year 2 means repaying principal out of capital rather than out of cash flow. That is not a financing problem you can solve with a better interest rate.

The generalisable rule: match the amortisation profile to the cash-generation profile. If an asset takes three years to reach cash break-even, the moratorium should be about three years. A lender loses nothing by granting it — interest accrues and is serviced throughout — and gains a materially safer credit.

The corollary for a borrower is that negotiating capital should be spent on structure, not on price. In this deal the moratorium is worth roughly 1.2x of DSCR. One hundred basis points of interest is worth about 0.03x.

The closed-form IDC solution

Interest during construction depends on the loan size, which depends on the project cost, which includes interest during construction. Most models iterate. The algebra is cleaner:

Solving the circularity

Let $k = d \times r \times (m \div 12) \times f$ where d is the debt proportion, r the rate, m the construction months and f the average drawdown factor.

Then $IDC = k \times (\text{HardCost} + IDC)$, which rearranges to:

$$IDC = k \times \text{HardCost} \div (1 - k)$$

Here $k = 0.70 \times 10.75\% \times 1.5 \times 0.45 = 0.05079$, giving IDC of Rs 563 Lakh on a hard cost of Rs 10,515 Lakh.

This matters practically. Enabling iterative calculation in Excel to resolve one intentional circularity also silences every unintentional one, which is how genuine errors survive review.

4.4 DSCR — and the two ways it is computed

Debt service coverage ratio is the single most important number in project finance, because it answers the lender's actual question: is there enough cash this year to make this year's payments? Two formulations are in common use and they are not the same:

Basis	Numerator	Used by	Comment
Cash	EBITDA – tax – increase in working capital	Project finance, infrastructure	Better — captures working capital absorption, which is what actually starves a ramping business of cash
Accrual	PAT + depreciation + interest	Most Indian sanction letters	Simpler, and ignores working capital entirely — which flatters a business with growing receivables

This model computes both. The workbook reports the cash basis as the primary measure and the accrual basis alongside it, so that the figures a lender computes from a sanction letter can be reconciled.

Three conventions matter and are frequently got wrong. First, DSCR is tested only in years when principal falls due — during a moratorium there is no meaningful ratio, and reporting a spuriously high or negative number for those years is a common presentation error. Second, the denominator uses term-loan interest and principal, not working-capital interest, which is a revolving facility rather than debt service. Third, the minimum matters more than the average: a lender is repaid or not repaid in a particular year, and an average of 1.66x is no comfort if Year 4 is 0.9x.

4.5 IRR, NPV and the difference between them

Measure	Question it answers	This project	Weakness
Project IRR	What return does the asset earn, before financing?	22.13%	Assumes interim cash flows are reinvested at the IRR itself
Equity IRR	What return does the shareholder earn, after debt service?	26.91%	Same, and highly sensitive to the exit assumption
NPV	How much value is created, in money?	Rs 123.19 Cr	Entirely dependent on the discount rate chosen
Money multiple	How many times is the cash returned?	12.3x	Ignores time completely
Payback	How long until the money comes back?	—	Ignores everything after payback

Use them together, because each is blind in a different direction. IRR is scale-blind: a 40% return on Rs 1 Cr beats a 25% return on Rs 100 Cr on IRR and loses badly on NPV. NPV is discount-rate-dependent: change WACC by two points and the answer moves materially, while IRR does not move at all. The sensitivity grid on exit multiple against WACC in the workbook demonstrates this directly — the IRR rows respond to the multiple and the columns are perfectly flat across WACC, which is a useful check that a model is wired correctly.

Why equity IRR exceeds project IRR here

Project IRR 22.13%. Post-tax cost of debt 8.04%. The asset out-earns the debt by roughly 14.1%, and leverage hands that spread to the shareholder — so equity IRR rises to 26.91%.

The identical mechanism works in reverse. When the asset return falls below the cost of debt, leverage subtracts from equity returns rather than adding, and it does so faster. That is exactly what happens in the pessimistic scenario, where no meaningful equity IRR exists at all.

And this is why the Funding Comparison sheet finds that de-levering costs almost nothing: once the loss-making ramp years are weighted in, the effective spread over the cost of debt is thin, so gearing is doing less work than the headline numbers suggest. The promoter is accepting a large increase in default probability for roughly 0.3 percentage points of return.

4.6 Sensitivity, tornado and Monte Carlo — three different tools

Technique	What it does	Strength	Blind spot
Scenario analysis	A few internally consistent combinations of assumptions	Coherent, easy to explain to a board	Says nothing about likelihood; the scenarios are chosen, not derived
One-way / tornado	Moves one driver at a time, ranks by impact	Identifies where management attention belongs	Real adversity arrives correlated, not one variable at a time
Two-way grid	Moves two drivers together across a surface	Shows interaction and locates covenant boundaries	Only two dimensions; everything else held at base
Monte Carlo	All drivers vary simultaneously from distributions	The only technique that produces probabilities	Output quality is entirely limited by the assumed distributions

These are complements, not substitutes, and the sequence matters. Use the tornado first to find which drivers deserve attention. Use two-way grids on the top pair to find where the covenant actually breaks. Use scenarios to tell a coherent story to a board. Use Monte Carlo last, to attach probabilities to the story.

Reading the tornado properly

The ranking is the output, not the values. In this project the four largest drivers of equity IRR are occupancy ramp speed, ARPOB escalation, ARPOB level and peak occupancy — every one demand-side, each swinging the return by 11 to 14 percentage points. Interest rate ranks eleventh of twelve at about 1.5 points.

That single ranking should reorganise a promoter's priorities. The instinct is to negotiate hardest on the loan, because pricing is the most visible negotiable number in the deal. The analysis says four percentage points of occupancy is worth more than three hundred basis points of interest. Management time belongs with clinicians, referrers and payers.

What Monte Carlo adds, and what it cannot

Running the model 10,000 times with every driver drawn simultaneously converts "this could happen" into "this happens 27% of the time". That is a genuinely different kind of statement, and it is what allows the Viability Report to size the standby facility off a distribution of cash deficits rather than off a single scenario.

One modelling detail worth copying: occupancy and ARPOB are drawn with a +0.35 correlation, because in practice stronger demand travels with a richer case mix. Treating them as independent would understate both tails and produce a comfortingly narrow distribution that does not exist.

Be sceptical of your own Monte Carlo

A simulation is the most persuasive output in any model and therefore the most dangerous. Four honest caveats apply to the one in this case study, and to almost every one you will encounter:

The distributions are judgemental. They are not fitted to observed data on comparable greenfield hospitals, because no such public dataset exists at this granularity. Precision in the output is not evidence of accuracy in the input.

Only one correlation is modelled. In reality wage inflation, consumable inflation and interest rates co-move, and all three co-move with demand. Modelling those would fatten the left tail considerably.

Triangular distributions have no tails past their bounds. Genuine catastrophe — a clinical scandal, a licence suspension, losing three anchor consultants at once — is entirely absent from these numbers. The real probability of

losing the equity is higher than any figure reported.

Execution quality is not a variable. The model assumes competent management throughout, which on the sector evidence is among the least safe assumptions in it.

Use the output comparatively: structure A is safer than structure B, by roughly this much. Do not use it as a forecast.

4.7 Valuation, and being honest about terminal value

Component	Gordon growth	Exit multiple
Sum of PV of forecast FCFF	Rs 65.28 Cr	Rs 65.28 Cr
Terminal value method	$FCFF_{10} \times (1+g) \div (WACC - g)$, $g = 5.0\%$	Year-10 EBITDA $\times$ 13.0x
Enterprise value	Rs 190.12 Cr	Rs 233.97 Cr
Terminal value as % of enterprise value	66%	72%

Between 66% and 72% of the answer lies beyond the forecast period. That is normal for a long-lived asset and it should still make you uncomfortable, because it means the valuation is mostly an assumption about perpetuity wearing the costume of a calculation.

Three habits help. Compute terminal value two independent ways and report both, as here — if Gordon growth and an exit multiple disagree wildly, at least one of them is wrong. Always disclose the terminal value share. And sensitise the terminal assumption explicitly rather than burying it: the grid in the workbook shows equity IRR moving from about 23% at 9x to about 30% at 17x, which is the honest width of the answer.

4.8 Cap tables and why greenfield rounds have no step-up

The PE Case sheet models institutional equity coming in alongside the promoter at financial closure. Shares are issued at cost — pre-money valuation equals the promoter's contributed capital and ownership is pro-rata to capital contributed.

That is not a modelling shortcut; it is what actually happens. A step-up requires something to have been de-risked, and at financial closure nothing has: no building, no licence, no patient, no clinician contract. The value created by the promoter up to that point is optionality — a site, a plan and a team — and a financial investor funding the construction risk will not pay a premium for it.

The step-up comes later. The defensible sequencing for a promoter is to close the debt now and raise institutional equity in Year 3 or 4 once occupancy is proven, at a genuine premium rather than at cost. That trades early risk-sharing for later, cheaper dilution. The deciding question is simply whether the promoter can fund the standby facility in the interim — which is a liquidity question, not a valuation one.

5. Indian-specific mechanics

Item	Treatment here	What to watch
Corporate tax	Section 115BAA: 22% + 10% surcharge + 4% cess = 25.168%	The concessional regime forgoes most incentives but removes MAT, which suits a loss-making ramp
Loss carry-forward	Business losses eight assessment years (s.72)	Unabsorbed depreciation carries forward indefinitely (s.32(2)) — a distinction that matters here
Book depreciation	Straight line, Companies Act 2013 Schedule II useful lives	Building 30 years, medical equipment 13, IT 6
Tax depreciation	Written down value, Income-tax Act rates	Building 10%, plant and machinery 15%, computers 40% — creates a deferred tax liability
Units	Rs Lakh throughout; 100 Lakh = 1 Crore	Indian project reports use Lakh or Crore, rarely millions. Be explicit in every header
GST	Healthcare services are largely exempt	Input tax credit is therefore mostly unavailable — a real cost that inflates equipment and consumables
Payer schemes	PMJAY, CGHS, ESI modelled at 72% realisation and 110 debtor days	Tariffs are administered and revised periodically; a revision is a genuine risk
Accreditation	NABH entry level then full, from Year 1	Insurance tariffs and many empanelments are conditional on it
Term loan conventions	9.5–12.5% floating, moratorium, quarterly instalments, DSCR 1.25x, TOL/TNW 2.5x	Personal guarantees are near-universal for promoter-driven hospital projects

This table is orientation, not authority. Rates, thresholds and scheme tariffs change. Verify against the Income-tax Act, Companies Act, GST notifications and current scheme circulars before relying on any of it.

6. Common modelling mistakes

Mistake	Why it happens	What to do instead
Growing revenue by a percentage	Fast, and looks reasonable	Build from drivers. Occupancy and ARPOB must exist as separate inputs to be sensitisable
No balance sheet check	The model "looks right"	Add the check row. It takes one formula and catches a whole class of error
Hardcoding a number mid-row	A one-off adjustment that never gets removed	Keep formulas identical across every projection period. A lone edited cell is the commonest silent error
Iterative calculation to fix circularity	It makes the error message disappear	Solve algebraically. Iteration also silences unintentional circular references
Reporting only the average DSCR	It looks better than the minimum	Report the minimum. A lender is repaid in a particular year, not on average
Testing DSCR during a moratorium	The formula runs in every column	Test only where principal falls due, and label the rest as moratorium
Equity value = EV – gross debt	Forgetting the cash belongs to shareholders	Use net debt. This exact error was found and fixed during verification of this model
Sensitising one variable at a time only	Tornados are easy and look thorough	Add correlated scenarios and a simulation. Real downside arrives bundled
Ignoring working capital in coverage	The accrual DSCR formula omits it	Use CFADS. Receivables are what actually starve a growing business of cash
Terminal value undisclosed	It is embarrassing when it is 70%	Disclose the share, use two methods, and sensitise the assumption
Sizing contingency off the base case	The base case is the plan	Size it off the distribution. The base case happens once
Presenting only the base case	It is the most persuasive	Lead with the risk. A reader who finds the downside themselves stops trusting the rest

7. Exercises

The workbook is live. These exercises change inputs and observe consequences — which is the only way to develop intuition for how a model behaves.

7.1 Structural

4. On the Control Panel, set the scenario to 3 (Pessimistic). Watch DSCR collapse and the cash balance turn deeply negative. Then work out from the Break-even sheet why the fall in EBITDA is so much larger, proportionally, than the fall in revenue.
5. Set the payer mix to 2 (Government-heavy). Track three numbers: the realisation index, debtor days, and minimum DSCR. Explain to yourself why the DSCR falls for two separate reasons rather than one.
6. On the Assumptions sheet, change the repayment ladder so the first instalment falls in Year 2 instead of Year 4 (move 4% into Year 2, keeping the total at 100%). Watch the minimum DSCR and the covenant verdict. This is the central finding of the case study, reproduced in about thirty seconds.
7. Set the funding structure to 3 (60:40) and compare equity IRR against the 70:30 case. It barely moves. Now compare the probability of technical default on the Funding Comparison sheet. Explain why the return is insensitive to gearing here when textbook theory says it should not be.

7.2 Analytical

8. Reduce peak occupancy until minimum DSCR touches exactly 1.25x. That occupancy is the project's true covenant floor — a far more useful number to a board than the base-case 72%.
9. Raise the consumables ratio from 23% to 26% and find how much extra occupancy is needed to restore the base-case EBITDA margin. This is the arithmetic of procurement discipline, and it is why it matters more than capex discipline.
10. On the Valuation sheet, set the exit multiple to 9.0x. Note how far equity IRR falls and how much of the enterprise value was resting on that single input.
11. Add nine months of construction delay via the Control Panel scenario. The IRR effect looks small. Explain why that is misleading, using the Scenario Summary sheet.

7.3 Extension

12. Build a Phase 2: 100 additional beds in Year 6, at a lower cost per bed because the land and support infrastructure already exist. Fund it from internal accruals plus a top-up loan. This is where the real value in a hospital platform lies, and this model deliberately excludes it — so the project as modelled understates the true opportunity.
13. Add a proper deferred tax schedule, tracking the timing difference between straight-line book depreciation and written-down-value tax depreciation, and carry a deferred tax liability on the balance sheet.
14. Make ALOS a variable rather than a constant, and separate it by specialty. Observe that shortening it raises ARPOB while reducing revenue per admission — and think about which one management should be incentivised on.
15. Model a competitor entering in Year 4: a 300 basis point occupancy loss and a 5% ARPOB reduction, sustained. Then ask whether the covenant survives it.

8. Glossary

Term	Definition
ALOS	Average length of stay — mean days per admission. $\text{Occupied bed-days} \div \text{admissions}$
ARPOB	Average revenue per occupied bed-day. $\text{Inpatient revenue} \div \text{occupied bed-days}$. Includes drugs and diagnostics consumed by inpatients
ARPP	Average revenue per patient. $\text{ARPOB} \times \text{ALOS}$
CFADS	Cash available for debt service. $\text{EBITDA less tax less the increase in working capital}$
COD	Commercial operation date — when the facility opens and revenue begins
Contribution margin	Revenue less variable costs. The amount available to cover fixed costs
DSCR	Debt service coverage ratio. $\text{CFADS} \div (\text{interest} + \text{principal})$. The central project finance test
DSRA	Debt service reserve account — ring-fenced cash held against future debt service
EBITDA	Earnings before interest, tax, depreciation and amortisation. The operating cash proxy
FCFF	Free cash flow to firm. Unlevered — before financing. Used for enterprise value
FCFE	Free cash flow to equity. After debt service. What the shareholder actually receives
IDC	Interest during construction — interest accrued before revenue begins, capitalised into the asset
IRR	Internal rate of return. The discount rate at which net present value equals zero
MOIC	Multiple on invested capital. $\text{Total cash returned} \div \text{cash invested}$. Ignores time
Moratorium	A period during which principal is not repaid. Interest is usually still serviced
NABH	National Accreditation Board for Hospitals and Healthcare Providers — the Indian quality standard
NPV	Net present value. Cash flows discounted at the cost of capital, less the investment
Occupancy	$\text{Occupied bed-days} \div (\text{beds} \times 365)$. The primary utilisation measure
PMJAY	Ayushman Bharat Pradhan Mantri Jan Arogya Yojana — India's government health insurance scheme
Realisation index	Weighted average of what each payer pays as a share of list tariff
ROCE	Return on capital employed. $\text{EBIT} \div (\text{net worth} + \text{borrowings})$
TOL/TNW	Total outside liabilities $\div$ tangible net worth. A leverage covenant, unreliable during a ramp
TPA	Third party administrator — processes claims between insurer and hospital
WACC	Weighted average cost of capital. The blended required return on debt and equity
WDV	Written down value — the reducing-balance depreciation method used by Indian tax law

9. What this model does badly

A companion that only listed strengths would be useless. These are the model's real weaknesses, in rough order of how much they matter:

16. The demand estimate is constructed, not researched. No patient-flow survey, physician interviews or competitor bed census underpins it. It is adequate for deciding whether to investigate further, not for committing Rs 110 Cr.
17. Phase 2 is entirely absent. The site carries FSI headroom and incremental beds on an existing structure are far cheaper than the first hundred. The model therefore understates the genuine option value of the project, possibly substantially.
18. Monte Carlo distributions are judgemental, single-correlation and bounded. Section 4.6 sets out why that matters and in which direction it biases the answer.
19. No competitor response is modelled. Incumbents will react to a new tertiary entrant, most plausibly on price and on consultant retention, and the model treats both as exogenous.
20. Tax is computed on book profit with no deferred tax schedule. Defensible for a case study, not for a filing or a real sanction.
21. ALOS is constant for ten years, which suppresses a genuine tension between clinical efficiency and case-mix enrichment.
22. The working capital facility is drawn at a fixed utilisation percentage rather than sized against the drawing power a lender would compute from stock and receivables.
23. Execution quality is not a variable. On sector evidence, management execution is among the most common causes of greenfield hospital underperformance, and it is nowhere in these numbers.

10. Closing thought

The most valuable output of this exercise is not the internal rate of return. It is the recognition that a project can be simultaneously worth building and badly financed, and that only structured risk analysis separates the two.

Fungi Labs has an attractive base case: Rs 110.78 Cr invested, a 22.1% project return, a 26.9% equity return, and value created of roughly Rs 101.27 Cr. A promoter reading only that would proceed with confidence. It also has a 27% probability of technical default and a covenant that holds in only 51% of simulated outcomes. A lender reading only that would decline.

Both readings are incomplete, and the analysis that reconciles them is what turns a spreadsheet into an appraisal. The answer was not to abandon the project or to accept it as proposed. It was to find the one structural change — deferring the first principal instalment by two years — that converted an unbankable proposal into a bankable one without altering the leverage, the pricing or the lender's yield.

That is the habit worth taking from this case study. Build the model to be interrogated rather than admired. Lead with what could go wrong. And when the numbers and the structure disagree, fix the structure.

This Learning Companion is document 4 of 4, accompanying the Feasibility Study, Detailed Project Report, Viability & Bankability Report, master financial model workbook and investor presentation. Fungi Labs is a fictitious entity; everything here is an illustrative case study and not professional advice.